

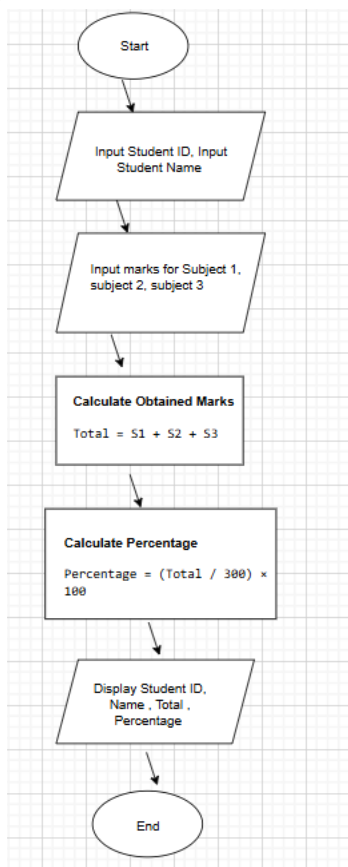
Objectives of the Lab

The objectives of this lab are:

1. To provide students with a basic understanding of fundamental programming concepts.
2. To enhance problem-solving skills using simple, real-world problems.
3. To introduce the Problem Analysis Chart (PAC) method for structured problem analysis.
4. To familiarize students with the Input-Process-Output (IPO) model.
5. To develop the ability to design and interpret flowcharts for program logic.
6. To understand the purpose and importance of pseudocode in program design.
7. To learn commonly used keywords and operations in pseudocode.
8. To develop skills in writing clear and logical algorithms.
9. To gain hands-on experience with Scratch programming.

Problem # 1: Student Details Record System

A) Flowchart



B) ALGORITHM

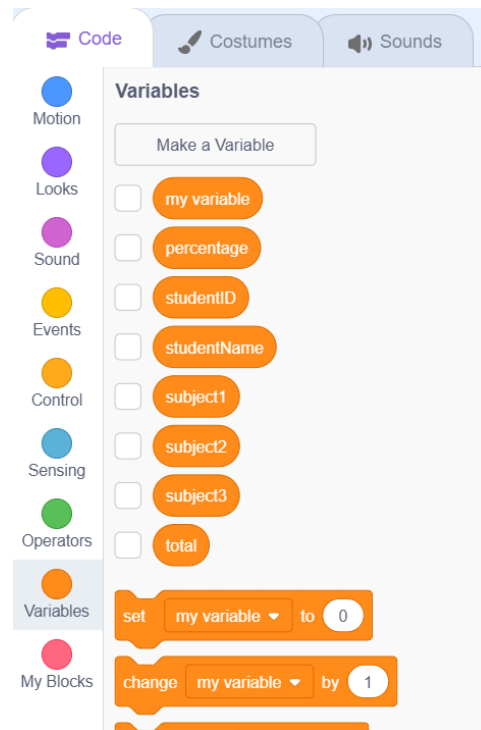
Algorithm: Student Record System

1. Start
2. Input Student ID
3. Input Student Name
4. Input marks of Subject 1
5. Input marks of Subject 2
6. Input marks of Subject 3
7. Calculate Obtained Marks
$$\text{Total} = \text{Subject1} + \text{Subject2} + \text{Subject3}$$
8. Calculate Percentage
$$\text{Percentage} = (\text{Total} / 300) \times 100$$
9. Display Student ID, Student Name, Obtained Marks, and Percentage
10. End

C) SCRATCH PROGRAM

STEP 1: Create Required Variables

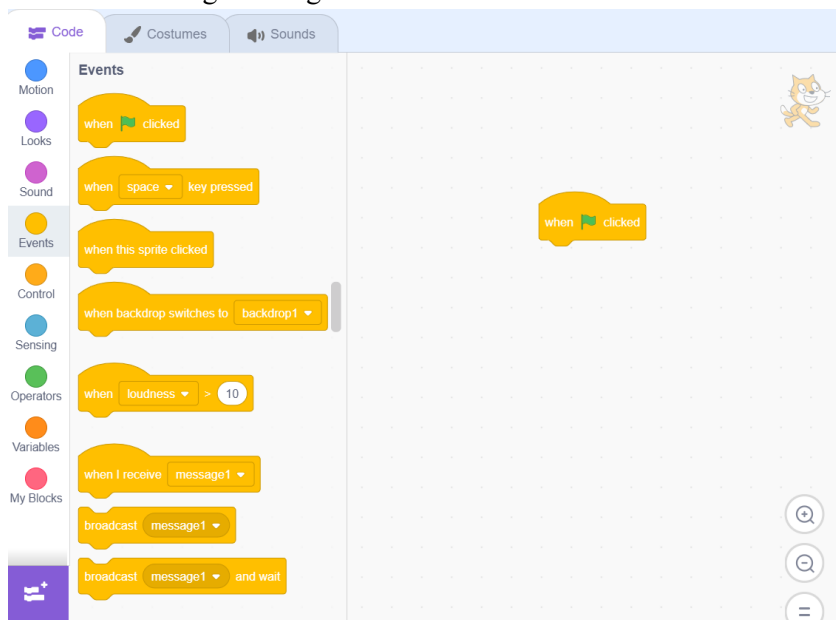
- On scratch Click on Variables and create the following variables:
 - studentID
 - studentName
 - subject1
 - subject2
 - subject3
 - total
 - percentage



This step creates all the required variables that will be used to store student details, subject marks, total marks, and percentage in the program.

STEP 2: Start the Program

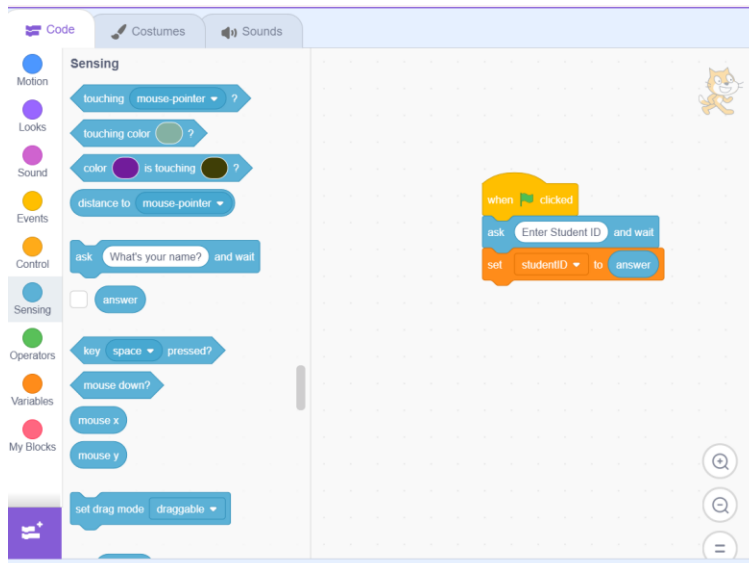
- Events → when green flag clicked



This block starts the program when the green flag is clicked.

STEP 3: Ask for Student ID

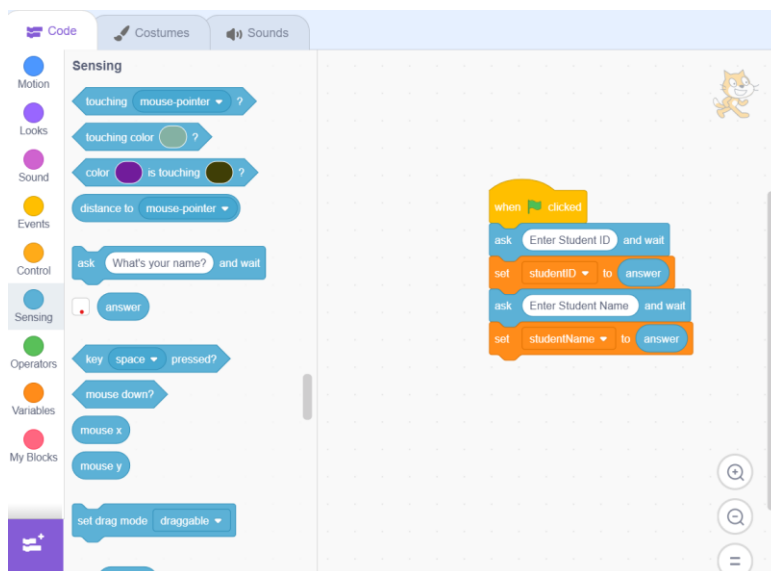
- Sensing → ask "Enter Student ID" and wait
- Variables → set studentID to answer



“This step asks the user to enter the student ID and stores the input in the variable studentID.”

STEP 4: Ask for Student Name

- ask "Enter Student Name" and wait
- set studentName to answer

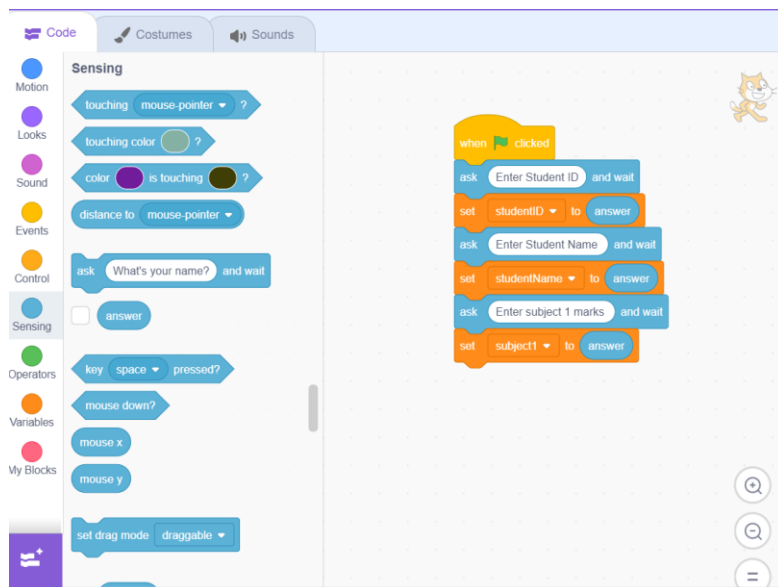


“This step takes the student’s name from the user and saves it in the variable.”

STEP 5: Input Subject Marks

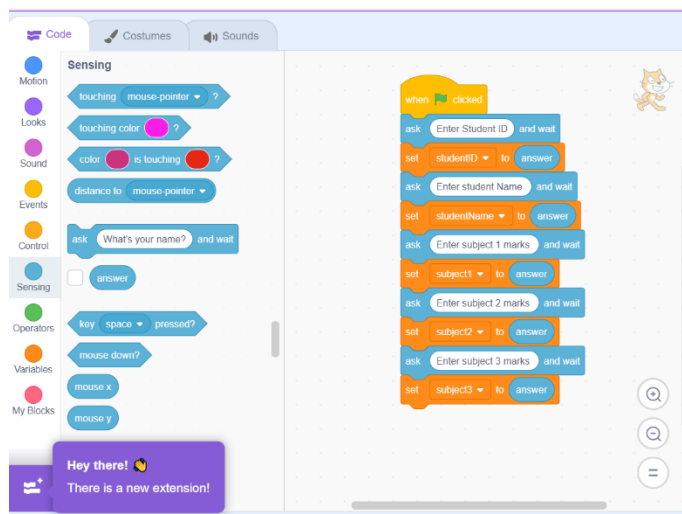
Subject 1

- ask "Enter Subject 1 marks" and wait
- set subject1 to answer



“This block asks for Subject 1 marks and stores them in the variable subject1.”

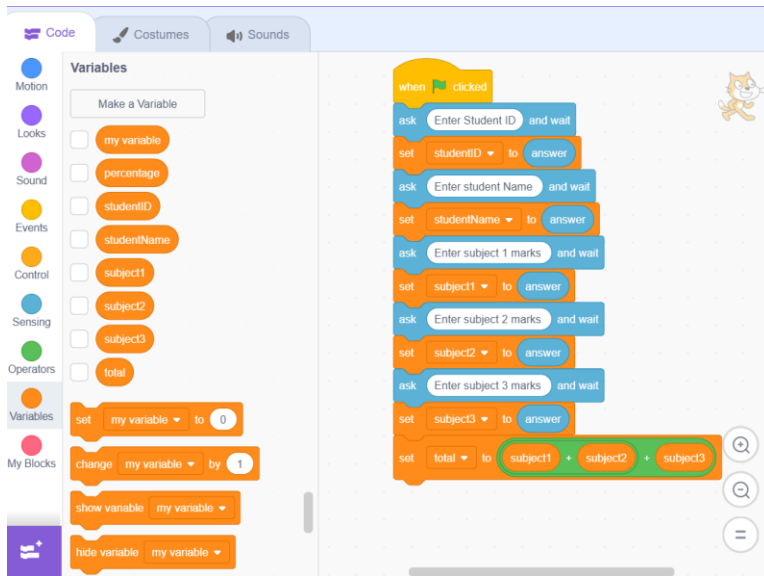
Subject 2 & Subject 3 (same pattern)



“These blocks collect marks for Subject 2 and Subject 3 and store them in their respective variables.”

STEP 6: Calculate Obtained Marks

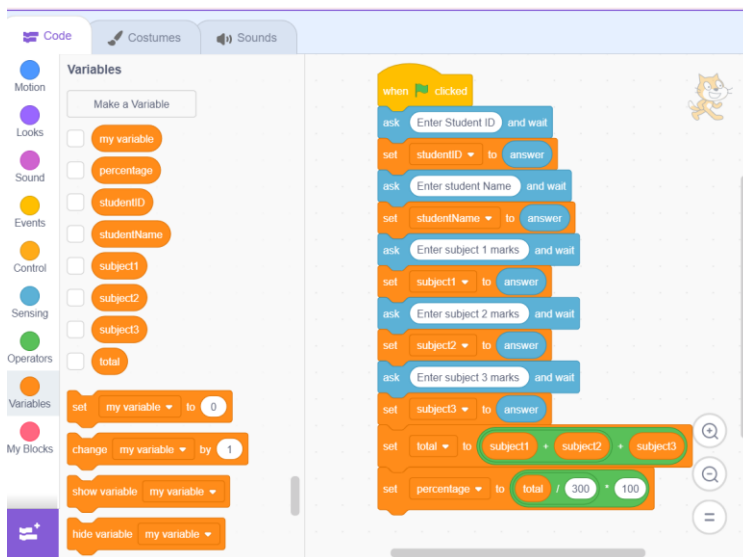
- set total to subject1 + subject2 + subject3



“This step adds all three subject marks and stores the total obtained marks in the variable total.”

STEP 7: Calculate Percentage

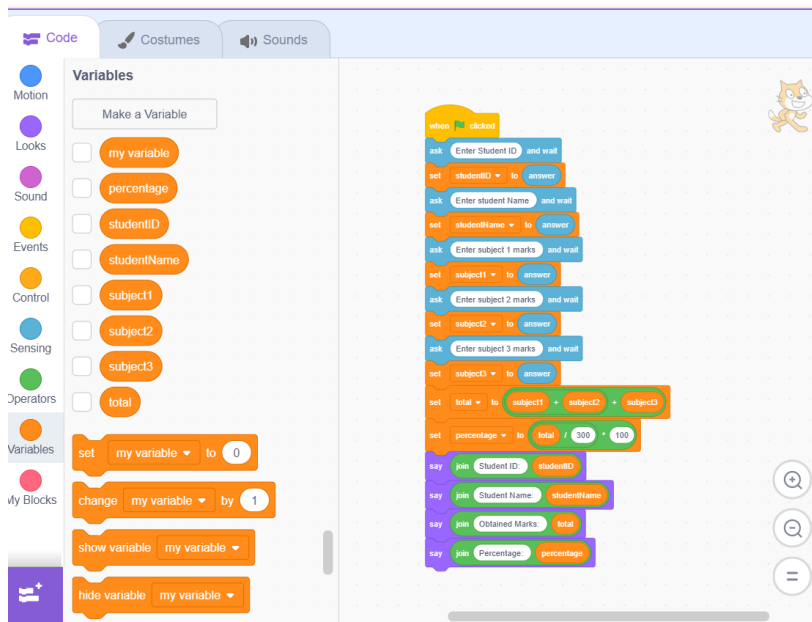
- set percentage to $(\text{total} / 300) * 100$



This block calculates the percentage based on total marks out of 300.”

STEP 8: Display Final Result

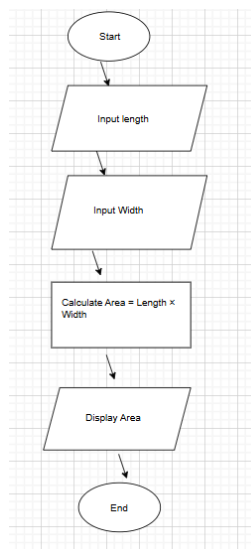
- say join "Student ID: " studentID
- say join "Student Name: " studentName
- say join "Obtained Marks: " total
- say join "Percentage: " percentage



This step displays the student's ID, name, total marks, and percentage.”

Problem # 2: Area of Rectangle

A) FLOWCHART



B) ALGORITHM

Algorithm: Area of Rectangle

1. Start
2. Input the length of the rectangle
3. Input the width of the rectangle

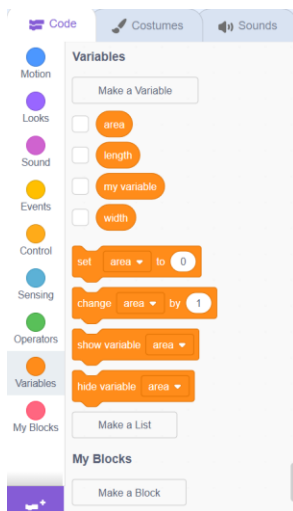
4. Calculate area = length $\times$ width
5. Display the calculated area
6. End

C) SCRATCH PROGRAM

STEP 1: Create Required Variables

On scratch Click on Variables and create the following variables:

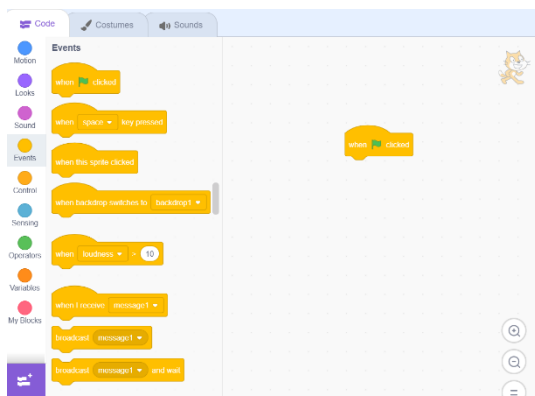
- length
- width
- area



“This step creates variables to store the length, width, and calculated area of the rectangle.”

STEP 2: Start the Program

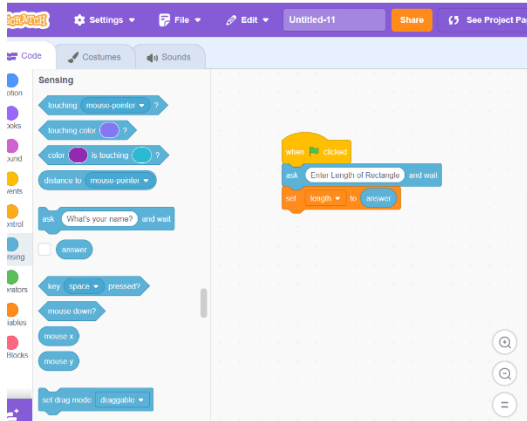
- Events $\rightarrow$ when green flag clicked



“This block starts the program when the green flag is clicked.”

STEP 3: Ask for Length

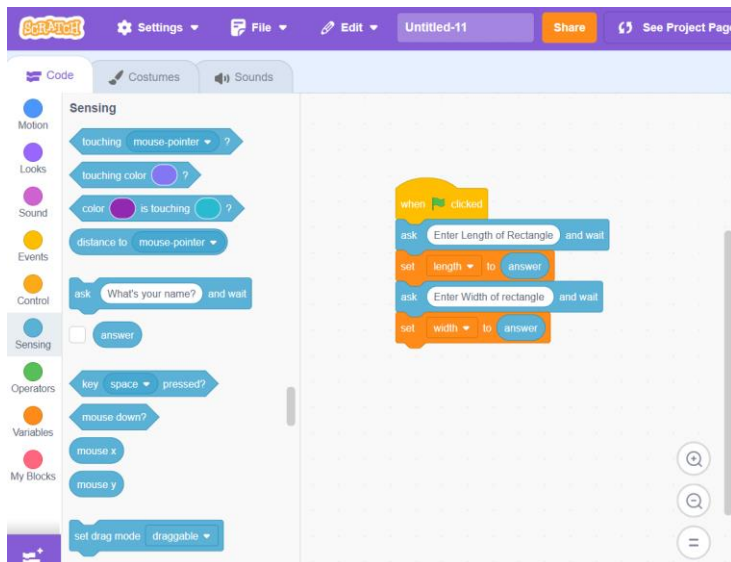
- ask "Enter length of rectangle" and wait
- set length to answer



“This step asks the user to enter the length of the rectangle and stores it in the variable length.”

STEP 4: Ask for Width

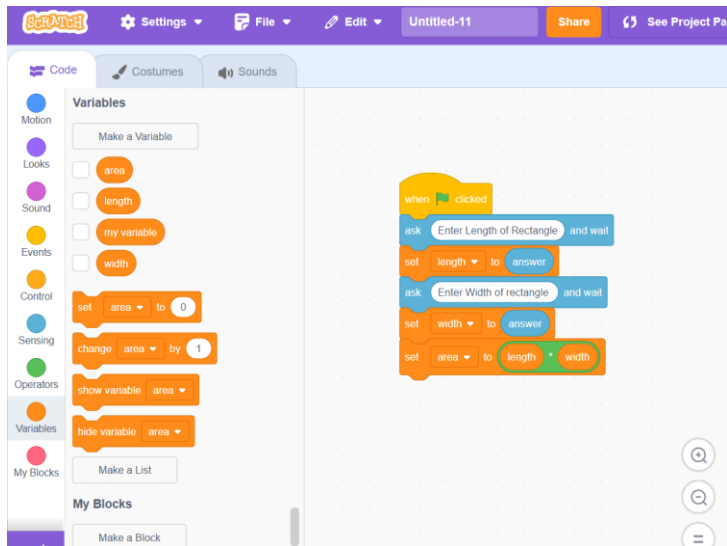
- ask "Enter width of rectangle" and wait
- set width to answer



“This step asks the user to enter the width of the rectangle and stores it in the variable width.”

STEP 5: Calculate Area

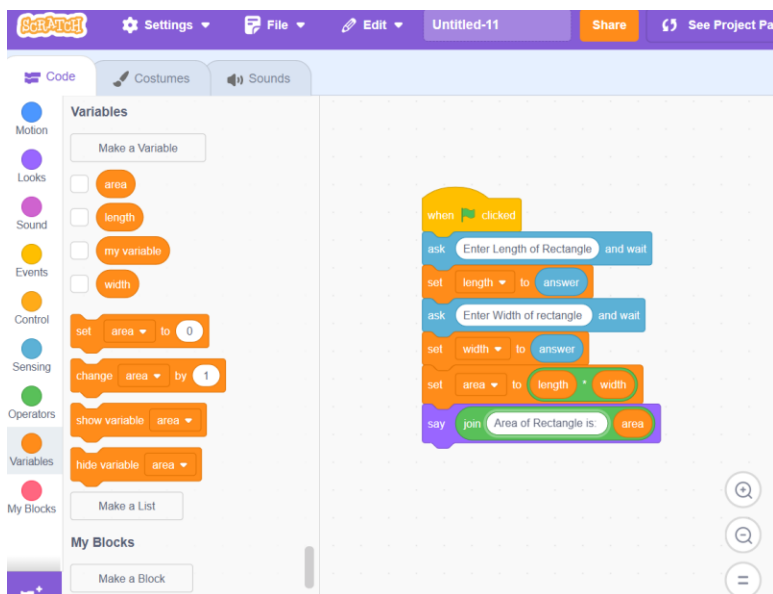
- set area to length * width



“This step calculates the area of the rectangle by multiplying the length and width.”

STEP 6: Display Area

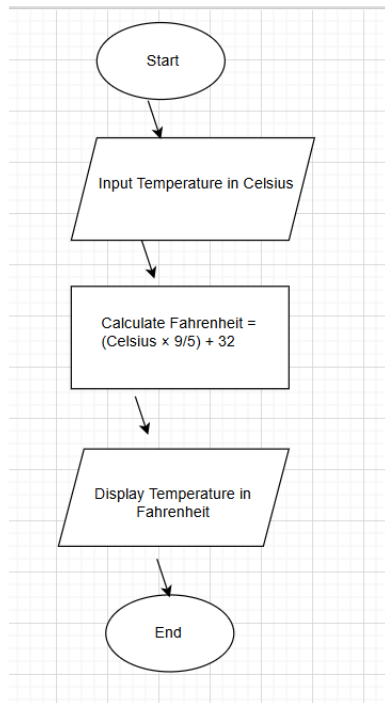
- say join "Area of rectangle is: " area



“This block displays the calculated area of the rectangle.”

Problem # 3: Celsius to Fahrenheit Conversion

A) FLOWCHART



B) ALGORITHM

Algorithm: Celsius to Fahrenheit Conversion

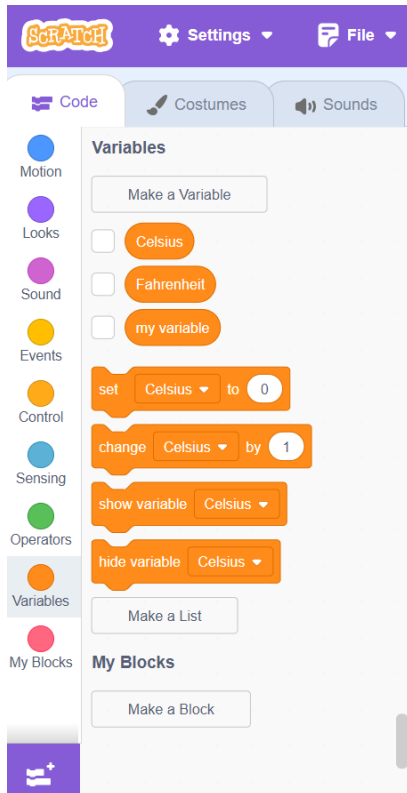
1. Start
2. Input temperature in Celsius
3. Calculate Fahrenheit = $(\text{Celsius} \times 9/5) + 32$
4. Display the temperature in Fahrenheit
5. End

C) SCRATCH PROGRAM

STEP 1: Create Required Variables (MANDATORY)

Variables to create:

- Celsius
- Fahrenheit

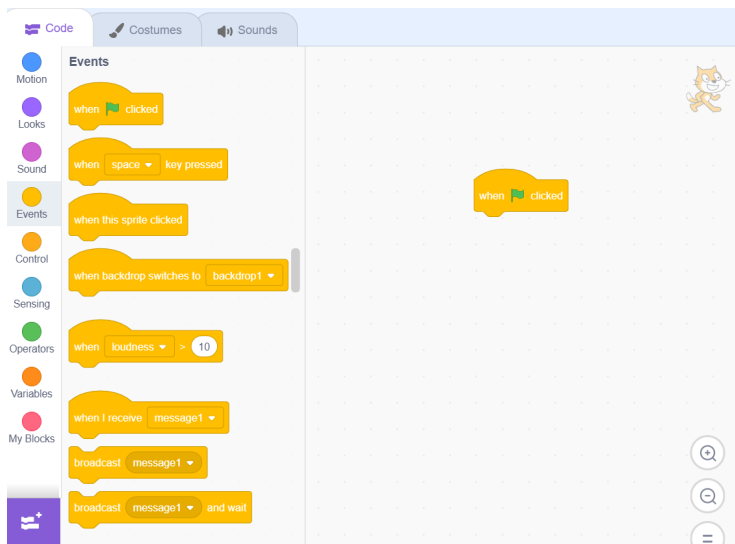


“This step creates variables to store the temperature in Celsius and the converted temperature in Fahrenheit.”

STEP 2: Start the Program

Block used:

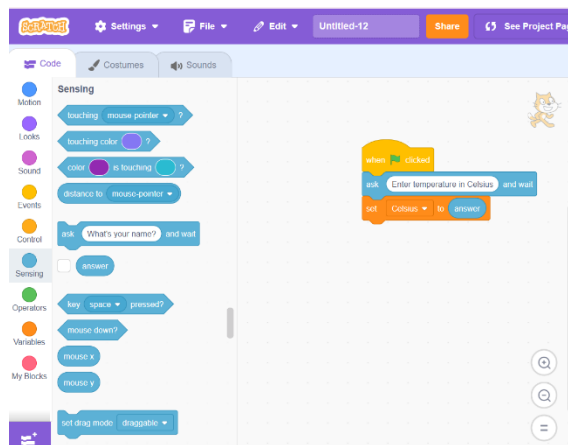
Events → when green flag clicked



“This block starts the temperature conversion program when the green flag is clicked.”

STEP 3: Ask for Celsius Temperature

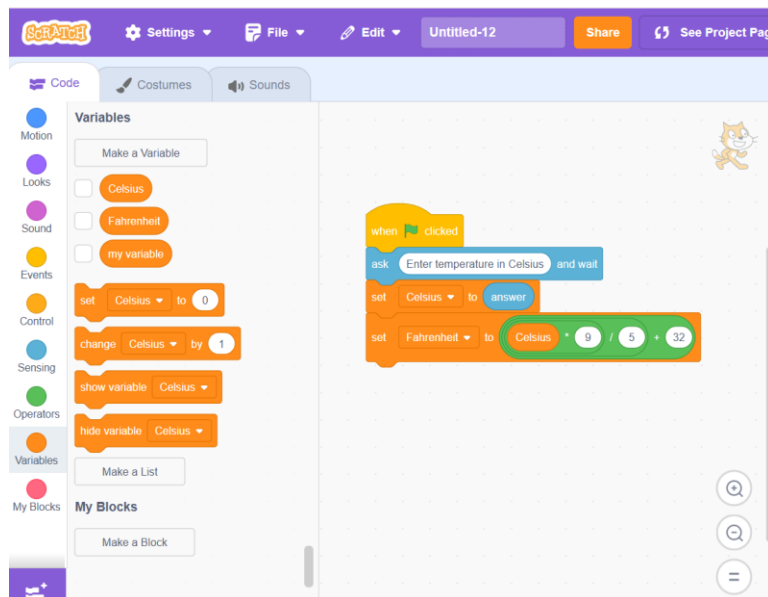
- ask "Enter temperature in Celsius" and wait
- set celsius to answer



“This step asks the user to enter the temperature in Celsius and stores it in the variable Celsius.”

STEP 4: Convert Celsius to Fahrenheit

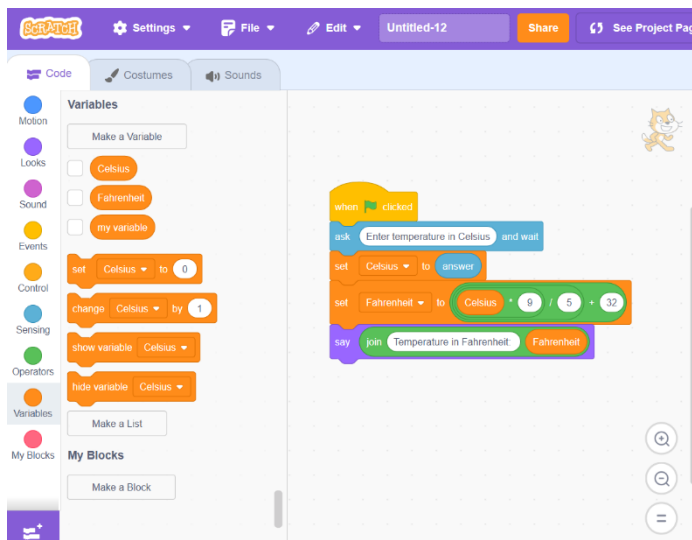
- set Fahrenheit to $(\text{Celsius} * 9 / 5) + 32$



“This step converts the Celsius temperature into Fahrenheit using the given formula.”

STEP 5: Display Fahrenheit Temperature

- say join "Temperature in Fahrenheit: " Fahrenheit



“This block displays the converted temperature in Fahrenheit.”

